MOTECH AUDIT

SMART CONTRACT SECURITY AUDIT

SECURE TODAY
BUILD TOMORROW

SMART CONTRACT

SECURITY AUDIT REPORT

PROJECT NAME

D3 TOKEN AUDIT REPORT

CONTRACT FILE	D3Token.sol
BLOCKCHAIN	BNB CHAIN
AUDIT DATE	20 OCT 2026
REPORT VERSION	v2.0

MOTECH AUDIT
Professional Smart Contract Security Audit Service
www.motechaudit.com

SECURITY ASSESSMENT

SUMMARY

This report has been prepared for D3 Token smart contracts, to discover issues and vulnerabilities in the source code of their Smart Contract as well as any contract dependencies that were not part of an officially recognise ed library. A comprehensive examination has been performed, utilising Static Analysis and ManualReview techniques.

The auditing process pays special attention to the following considerations:

- Testing the smart contracts against both common and uncommon attack vectors.
- Assessing the codebase to ensure compliance with current best practices and industry standards.
- Ensuring contract logic meets the specifications and intentions of the client.
- Cross referencing contract structure and implementation against similar smart contracts produced by industry leaders.
- Thorough line-by-line manual review of the entire codebase by industry experts.

The security assessment resulted in findings that ranged from critical to informational. We recommend addressing these findings to ensure a high level of security standards and industry practices. We suggest recommendations that could better serve the project from the security perspective:

- Enhance general coding practices for better structures of source codes;
- Add enough unit tests to cover the possible use cases given they are currently missing in the repository;
- Provide more comments per each function for readability, especially contracts are verified in public;
- Provide more transparency on privileged activities once the protocol is live.

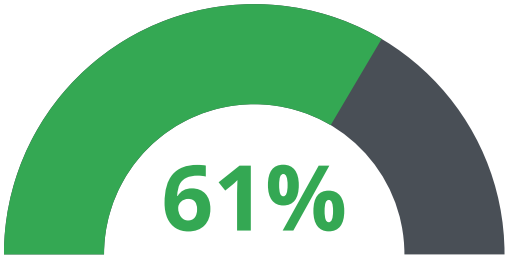
SECURITY ASSESSMENT

SUMMARY

Pass/Fail	Analyzer	Description
✓	Motech Audit	Vulnerability analyzer built by Motech Audit that quickly searches within smart contracts for vulnerable code fragments not only exact but also syntactically different clones.
⚠	Formal Verifier	Sound formal verifier built by Motech Audit that analyzes within smart contracts for proving that a program satisfies nonexistence of vulnerability such as integer overflow.
✗	Achilles	Symbolic analyzer built by Motech Audit that extracts code patterns within smart contracts for finding syntactic and semantic bugs and vulnerabilities based on SWC specification.

✓ Pass ⚠ Compile Error ✗ Failed

SECURITY SCORE



File : D3Token.sol
Language : solidity
Size. : 23050 bytes
Date : 2026-10-20T12:41:16.281Z

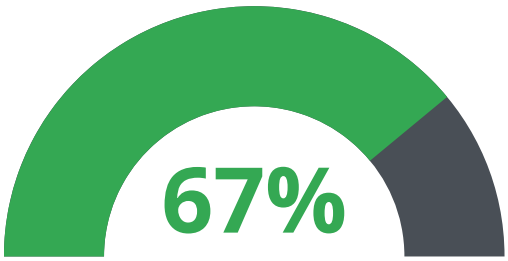
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SECURITY ASSESSMENT

SUMMARY

THREAT SCORE



High Risk	00
Moderate Risk	01
Low Risk	03
No Impact	18
Beneficial	07
Unavailable	10



Token is NOT Honeypot

The token contract is not a honeypot. Code shows standard buy/sell tax routing to marketWallet and a time-gated deflation that burns a computed fraction (DEFLATION_BPS) from mainPair. No allowance bypass, hidden privileged mappings, arbitrary balance rewrites, minting, or conditional zeroing are present in the provided snippets.



SECURITY ASSESMENT

DISCLAIMER

This is a limited report on our findings based on our analysis, in accordance with good industry practice as at the date of this report, in relation to cybersecurity vulnerabilities and issues in the framework and algorithms based on smart contracts, the details of which are set out in this report. In order to get a full view of our analysis, it is crucial for you to read the full report. While we have done our best in conducting our analysis and producing this report, it is important to note that you should not rely on this report and cannot claim against us on the basis of what it says or doesn't say, or how we produced it, and it is important for you to conduct your own independent investigations before making any decisions. We go into more detail on this in the below disclaimer below – please make sure to read it in full.

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The analysis of the security is purely based on the smart contracts alone. No applications or operations were reviewed for security. No product code has been reviewed.

SECURITY ASSESSMENT

BACKGROUND

Motech Audit was commissioned by D³ Finance to perform an audit of smart contracts:

D3Token.sol
4bcdfc70a4e408271847f12f315b81446ccad52ad9aa7d1d3201a73992fcf7c2

The purpose of the audit was to achieve the following:

- Ensure that the smart contract functions as intended.
- Identify potential security issues with the smart contract.

The information in this report should be used to understand the risk exposure of the smart contract, and as a guide to improve the security posture of the smart contract by remediating the issues that were identified.

SECURITY ASSESSMENT

VULNERABILITY & RISK LEVEL

58 Total Issues

Risk represents the probability that a certain source-threat will exploit vulnerability, and the impact of that event on the organization or system. Risk Level is computed based on CVSS version 3.0.

Level	Total Issues	Vulnerability	Risk (Required Action)
Critical	0	A vulnerability that can disrupt the contract functioning in a number of scenarios,or creates a risk that the contract may be broken	Immediate action to reduce risk level.
High	1	A vulnerability that affects the desired outcome when using a contract, or provides the opportunity to use a contract in an unintended way.	Implementation of corrective actions as soon as possible.
Medium	17	A vulnerability that could affect the desired outcome of executing the contract in a specific scenario.	Implementation of corrective actions in a certain period.
Low	14	A vulnerability that does not have a significant impact on possible scenarios for the use of the contract and is probably subjective.	Implementation of certain corrective actions or accepting the risk.
Note	26	A vulnerability that have informational character but is not affecting any of the code.	An observation that does not determine a level of risk.

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SECURITY ASSESSMENT

AUDITING STRATEGY AND TECHNIQUES APPLIED

Throughout the review process, care was taken to evaluate the repository for security-related issues, code quality, and adherence to specification and best practices. To do so, reviewed line-by-line by our team of expert pentesters and smart contract developers, documenting any issues as there were discovered.

Methodology

The auditing process follows a routine series of steps:

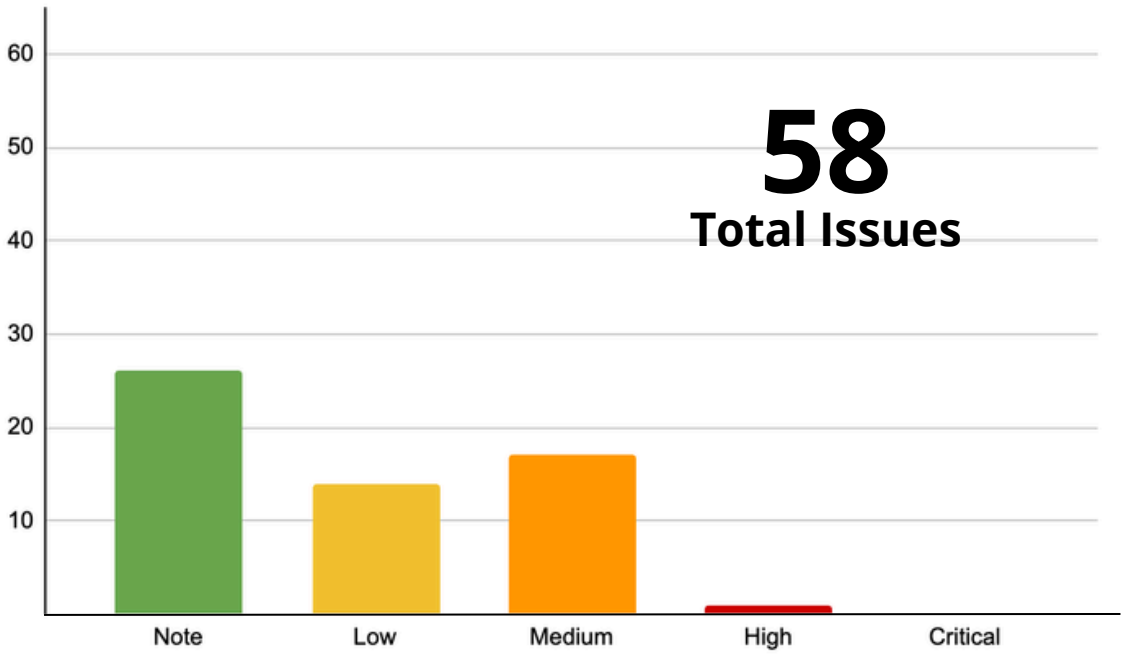
1. Code review that includes the following:
- Review of the specifications, sources, and instructions provided to Motech Audit to make sure we understand the size, scope, and functionality of the smart contract.
- Manual review of code, which is the process of reading source code line-byline in an attempt to identify potential vulnerabilities.
- Comparison to specification, which is the process of checking whether the code does what the specifications, sources, and instructions provided to Motech Audit describe.

2. Testing and automated analysis that includes the following:
- Test coverage analysis, which is the process of determining whether the test cases are actually covering the code and how much code is exercised when we run those test cases.
- Symbolic execution, which is analysing a program to determine what inputs causes each part of a program to execute.

3. Best practices review, which is a review of the smart contracts to improve efficiency, effectiveness, clarify, maintainability, security, and control based on the established industry and academic practices, recommendations, and research.

4. Specific, itemized, actionable recommendations to help you take steps to secure your smart contracts.

SECURITY ASSESSMENT
FINDINGS



Audit Score Summary

Category	Score (Out of 10)
Code Correctness	7.0
Security	5.0
Upgrade Safety	7.0
Access Control	6.0
Gas Efficiency	7.0
Documentation	6.0
Test Coverage	4.0
Overall Audit Score	61 / 100

SECURITY ASSESSMENT

PROJECT OVERVIEW

D3Token is an ERC-20-compatible token implemented using OpenZeppelin components, including ERC20, ERC20Burnable, ERC20Permit, Ownable, and AccessControl. The contract incorporates a fixed initial token supply of 21,000,000 D3 tokens, transaction taxation, buy-side trading controls, dynamic sell-tax mechanisms, TWAP-related price monitoring, liquidity-pair interactions, and a periodic deflation mechanism. The declared initial allocation consists of 60% for the interaction pool, 20% for the emission pool, 15% for the incentive pool, and 5% for the liquidity pool. The reviewed source code does not contain a separate post-deployment minting function.

The contract applies a base buy tax of 3% and a base sell tax of 3%, with additional sell-tax mechanisms associated with price deviation and liquidity-pool value. The declared maximum total sell tax is 30%, meaning that the effective tax rate may vary depending on the applicable conditions and contract configuration. The contract also incorporates TWAP-related price observations using fast and slow time windows to support price-deviation assessment and related tax calculations. The reliability of these mechanisms depends on correct configuration, observation freshness, liquidity conditions, and the behavior of the external liquidity pair.

The revised implementation introduces several positive security and design improvements. The emergency withdrawal functions have been removed, reducing the scope for privileged asset extraction through those specific mechanisms, while the dedicated sell-disable mechanism has also been removed and replaced with a buy-side control. In addition, the deflation mechanism has been redesigned to apply a fixed 0.2% burn of the D3 token balance held by the configured main liquidity pair, with execution limited to once per UTC day. The burned tokens are transferred to the zero address, and the contract invokes the liquidity pair's sync() function to update its reserve accounting. This approach provides a more clearly defined deflation process and directly reduces the token's

SECURITY ASSESSMENT

PROJECT OVERVIEW

Despite these improvements, certain privileged configuration powers remain. The owner can modify tax-exempt addresses and change the configured main liquidity pair, while the owner or administrator can disable the deflation mechanism. The maximum total sell tax of 30% may materially affect users' exit costs, and the interaction between tax exemptions, buy-side restrictions, dynamic sell taxation, oracle observations, and liquidity-pair identification remains relevant to trading consistency and user protection. Token allocation concentration and the privileges assigned to project-associated addresses should also be considered when evaluating the contract's overall security and operational transparency.

The project team reports that 24 local Hardhat simulation scenarios passed, covering buy restrictions, sell availability, permission boundaries, daily deflation limits, token supply reduction, liquidity-pair synchronization, and a simulated buy-burn-sell arbitrage scenario. These results provide project-supplied testing information for the assessment but should not be represented as independently reproduced Motech Audit test results unless the corresponding scripts and execution evidence have been independently verified.

Overall, the revised D3Token contract demonstrates meaningful improvements to its trading-control design and deflation mechanism, particularly through the removal of the previously identified emergency withdrawal and sell-disable mechanisms. Nevertheless, privileged administrative controls, tax configuration, liquidity-pair management, oracle reliability, and token distribution remain important considerations in evaluating the contract's overall security and operational risk.

VULNERABILITY FINDINGS

The entries below reproduce scanner results and provide general remediation guidance. Recommendations must be validated against the complete contract and its intended behavior.

FINDING 01 | incorrect-exp

High

Detector: incorrect-exp

Code location: Line 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184, 185, 186, 187, 188, 189, 190, 191, 192, 193, 194, 195, 196, 197, 198, 199, 200, 201, 202

Scanner description: Math.mulDiv(uint256,uint256,uint256) (node_modules/@openzeppelin/contracts/utils/math/Math.sol#123-202) has bitwise-xor operator ^ instead of the exponentiation operator **: - inverse = (3 * denominator) ^ 2 (node_modules/@openzeppelin/contracts/utils/math/Math.sol#184)

Recommended action: Review this detector result in context, validate its impact, implement a targeted fix where necessary, and add regression tests.

FINDING 02 | divide-before-multiply

Medium

Detector: divide-before-multiply

Code location: Line 495, 496, 497, 498, 499, 500, 501, 502, 503, 504, 505, 506, 507, 508, 509, 510, 511, 512, 513, 514, 515, 516, 517, 518, 519, 520, 521, 522, 523, 524, 525, 526, 527, 528, 529, 530, 531, 532, 533, 534

Scanner description: D3Token._computeTwap(uint256,bool) (uploads/d52437ee22094038b7eb7b15a761c25f_D3Token_1.sol#495-534) performs a multiplication on the result of a division: - averageUQ112x112 = cumulativeDelta / elapsed (uploads/d52437ee22094038b7eb7b15a761c25f_D3Token_1.sol#531) - quotePerD3Raw = (averageUQ112x112 * 1e18) >> 112 (uploads/d52437ee22094038b7eb7b15a761c25f_D3Token_1.sol#532)

Recommended action: Review this detector result in context, validate its impact, implement a targeted fix where necessary, and add regression tests.

FINDING 03 | divide-before-multiply

Medium

Detector: divide-before-multiply

Code location: Line 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184, 185, 186, 187, 188, 189, 190, 191, 192, 193, 194, 195, 196, 197, 198, 199, 200, 201, 202

Scanner description: Math.mulDiv(uint256,uint256,uint256) (node_modules/@openzeppelin/contracts/utils/math/Math.sol#123-202) performs a multiplication on the result of a division: - denominator = denominator / twos (node_modules/@openzeppelin/contracts/utils/math/Math.sol#169) - inverse *= 2 - denominator * inverse (node_modules/@openzeppelin/contracts/utils/math/Math.sol#192)

Recommended action: Review this detector result in context, validate its impact, implement a targeted fix where necessary, and add regression tests.

FINDING 04 | divide-before-multiply

Medium

Detector: divide-before-multiply

Code location: Line 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184, 185, 186, 187, 188, 189, 190, 191, 192, 193, 194, 195, 196, 197, 198, 199, 200, 201, 202

Scanner description: Math.mulDiv(uint256,uint256,uint256) (node_modules/@openzeppelin/contracts/utils/math/Math.sol#123-202) performs a multiplication on the result of a division: - denominator = denominator / twos (node_modules/@openzeppelin/contracts/utils/math/Math.sol#169) - inverse *= 2 - denominator * inverse (node_modules/@openzeppelin/contracts/utils/math/Math.sol#190)

Recommended action: Review this detector result in context, validate its impact, implement a targeted fix where necessary, and add regression tests.

FINDING 05 | divide-before-multiply

Medium

Detector: divide-before-multiply

Code location: Line 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184, 185, 186, 187, 188, 189, 190, 191, 192, 193, 194, 195, 196, 197, 198, 199, 200, 201, 202

Scanner description: Math.mulDiv(uint256,uint256,uint256) (node_modules/@openzeppelin/contracts/utils/math/Math.sol#123-202) performs a multiplication on the result of a division: - denominator = denominator / twos (node_modules/@openzeppelin/contracts/utils/math/Math.sol#169) - inverse = (3 * denominator) ^ 2 (node_modules/@openzeppelin/contracts/utils/math/Math.sol#184)

Recommended action: Review this detector result in context, validate its impact, implement a targeted fix where necessary, and add regression tests.

FINDING 06 | divide-before-multiply

Medium

Detector: divide-before-multiply

Code location: Line 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184, 185, 186, 187, 188, 189, 190, 191, 192, 193, 194, 195, 196, 197, 198, 199, 200, 201, 202

Scanner description: Math.mulDiv(uint256,uint256,uint256) (node_modules/@openzeppelin/contracts/utils/math/Math.sol#123-202) performs a multiplication on the result of a division: - denominator = denominator / twos (node_modules/@openzeppelin/contracts/utils/math/Math.sol#169) - inverse *= 2 - denominator * inverse (node_modules/@openzeppelin/contracts/utils/math/Math.sol#191)

Recommended action: Review this detector result in context, validate its impact, implement a targeted fix where necessary, and add regression tests.

FINDING 07 | divide-before-multiply

Medium

Detector: divide-before-multiply

Code location: Line 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184, 185, 186, 187, 188, 189, 190, 191, 192, 193, 194, 195, 196, 197, 198, 199, 200, 201, 202

Scanner description: Math.mulDiv(uint256,uint256,uint256) (node_modules/@openzeppelin/contracts/utils/math/Math.sol#123-202) performs a multiplication on the result of a division: - denominator = denominator / twos (node_modules/@openzeppelin/contracts/utils/math/Math.sol#169) - inverse *= 2 - denominator * inverse (node_modules/@openzeppelin/contracts/utils/math/Math.sol#188)

Recommended action: Review this detector result in context, validate its impact, implement a targeted fix where necessary, and add regression tests.

FINDING 08 | divide-before-multiply

Medium

Detector: divide-before-multiply

Code location: Line 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184, 185, 186, 187, 188, 189, 190, 191, 192, 193, 194, 195, 196, 197, 198, 199, 200, 201, 202

Scanner description: Math.mulDiv(uint256,uint256,uint256) (node_modules/@openzeppelin/contracts/utils/math/Math.sol#123-202) performs a multiplication on the result of a division: - prod0 = prod0 / twos (node_modules/@openzeppelin/contracts/utils/math/Math.sol#172) - result = prod0 * inverse (node_modules/@openzeppelin/contracts/utils/math/Math.sol#199)

Recommended action: Review this detector result in context, validate its impact, implement a targeted fix where necessary, and add regression tests.

FINDING 09 | divide-before-multiply

Medium

Detector: divide-before-multiply

Code location: Line 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184, 185, 186, 187, 188, 189, 190, 191, 192, 193, 194, 195, 196, 197, 198, 199, 200, 201, 202

Scanner description: Math.mulDiv(uint256,uint256,uint256) (node_modules/@openzeppelin/contracts/utils/math/Math.sol#123-202) performs a multiplication on the result of a division: - denominator = denominator / twos (node_modules/@openzeppelin/contracts/utils/math/Math.sol#169) - inverse *= 2 - denominator * inverse (node_modules/@openzeppelin/contracts/utils/math/Math.sol#193)

Recommended action: Review this detector result in context, validate its impact, implement a targeted fix where necessary, and add regression tests.

FINDING 10 | divide-before-multiply

Medium

Detector: divide-before-multiply

Code location: Line 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184, 185, 186, 187, 188, 189, 190, 191, 192, 193, 194, 195, 196, 197, 198, 199, 200, 201, 202

Scanner description: Math.mulDiv(uint256,uint256,uint256) (node_modules/@openzeppelin/contracts/utils/math/Math.sol#123-202) performs a multiplication on the result of a division: - denominator = denominator / twos (node_modules/@openzeppelin/contracts/utils/math/Math.sol#169) - inverse *= 2 - denominator * inverse (node_modules/@openzeppelin/contracts/utils/math/Math.sol#189)

Recommended action: Review this detector result in context, validate its impact, implement a targeted fix where necessary, and add regression tests.

FINDING 11 | divide-before-multiply

Medium

Detector: divide-before-multiply

Code location: Line 470, 471, 472, 473, 474, 475, 476, 477, 478, 479, 480, 481, 482, 483, 484, 485, 486, 487, 488, 489, 490, 491, 492

Scanner description: D3Token._currentPairState() (uploads/d52437ee22094038b7eb7b15a761c25f_D3Token_1.sol#470-492) performs a multiplication on the result of a division: - priceUQ112x112 = (reserveQuote << 112) / reserveD3 (uploads/d52437ee22094038b7eb7b15a761c25f_D3Token_1.sol#488) - currentCumulative += priceUQ112x112 * elapsed (uploads/d52437ee22094038b7eb7b15a761c25f_D3Token_1.sol#489)

Recommended action: Review this detector result in context, validate its impact, implement a targeted fix where necessary, and add regression tests.

FINDING 12 | incorrect-equality

Medium

Detector: incorrect-equality

Code location: Line 287, 288, 289, 290, 291, 292, 293, 294, 295, 296, 297, 298, 299

Scanner description: D3Token.gateTaxBps() (uploads/d52437ee22094038b7eb7b15a761c25f_D3Token_1.sol#287-299) uses a dangerous strict equality: - fastPrice == 0 || slowPrice == 0 || fastPrice >= slowPrice (uploads/d52437ee22094038b7eb7b15a761c25f_D3Token_1.sol#290)

Recommended action: Review this detector result in context, validate its impact, implement a targeted fix where necessary, and add regression tests.

FINDING 13 | incorrect-equality

Medium

Detector: incorrect-equality

Code location: Line 340, 341, 342, 343

Scanner description: D3Token.nextDeflationAt() (uploads/d52437ee22094038b7eb7b15a761c25f_D3Token_1.sol#340-343) uses a dangerous strict equality: - lastDeflationAt == 0 (uploads/d52437ee22094038b7eb7b15a761c25f_D3Token_1.sol#341)

Recommended action: Review this detector result in context, validate its impact, implement a targeted fix where necessary, and add regression tests.

FINDING 14 | incorrect-equality

Medium

Detector: incorrect-equality

Code location: Line 417, 418, 419, 420, 421, 422, 423, 424, 425, 426, 427, 428, 429, 430, 431, 432, 433, 434, 435, 436, 437, 438, 439, 440, 441, 442, 443, 444, 445, 446, 447, 448, 449, 450, 451, 452, 453, 454, 455

Scanner description: D3Token._updateOracle() (uploads/d52437ee22094038b7eb7b15a761c25f_D3Token_1.sol#417-455) uses a dangerous strict equality: - currentSpot == 0 (uploads/d52437ee22094038b7eb7b15a761c25f_D3Token_1.sol#421)

Recommended action: Review this detector result in context, validate its impact, implement a targeted fix where necessary, and add regression tests.

FINDING 15 | uninitialized-local

Medium

Detector: uninitialized-local

Code location: Line 504

Scanner description: D3Token._computeTwap(uint256,bool).earliest (uploads/d52437ee22094038b7eb7b15a761c25f_D3Token_1.sol#504) is a local variable never initialized

Recommended action: Review this detector result in context, validate its impact, implement a targeted fix where necessary, and add regression tests.

FINDING 16 | uninitialized-local

Medium

Detector: uninitialized-local

Code location: Line 503

Scanner description: D3Token._computeTwap(uint256,bool).referenceObservation (uploads/d52437ee22094038b7eb7b15a761c25f_D3Token_1.sol#503) is a local variable never initialized

Recommended action: Review this detector result in context, validate its impact, implement a targeted fix where necessary, and add regression tests.

FINDING 17 | uninitialized-local

Medium

Detector: uninitialized-local

Code location: Line 202

Scanner description: D3Token.setMainPair(address).quoteDecimals_ (uploads/d52437ee22094038b7eb7b15a761c25f_D3Token_1.sol#202) is a local variable never initialized

Recommended action: Review this detector result in context, validate its impact, implement a targeted fix where necessary, and add regression tests.

FINDING 18 | uninitialized-local

Medium

Detector: uninitialized-local

Code location: Line 505

Scanner description: D3Token._computeTwap(uint256,bool).fullWindowFound (uploads/d52437ee22094038b7eb7b15a761c25f_D3Token_1.sol#505) is a local variable never initialized

Recommended action: Review this detector result in context, validate its impact, implement a targeted fix where necessary, and add regression tests.

FINDING 19 | shadowing-local

Low

Detector: shadowing-local

Code location: Line 39, 58, 59, 60, 15

Scanner description: ERC20Permit.constructor(string).name (node_modules/@openzeppelin/contracts/token/ERC20/extensions/ERC20Permit.sol#39) shadows: - ERC20.name() (node_modules/@openzeppelin/contracts/token/ERC20/ERC20.sol#58-60) (function) - IERC20Metadata.name() (node_modules/@openzeppelin/contracts/token/ERC20/extensions/IERC20Metadata.sol#15) (function)

Recommended action: Review this detector result in context, validate its impact, implement a targeted fix where necessary, and add regression tests.

FINDING 20 | missing-zero-check

Low

Detector: missing-zero-check

Code location: Line 149, 162

Scanner description: D3Token.constructor(address,address,address,address,address,address,address).marketWallet_ (uploads/d52437ee22094038b7eb7b15a761c25f_D3Token_1.sol#149) lacks a zero-check on : - marketWallet = marketWallet_ (uploads/d52437ee22094038b7eb7b15a761c25f_D3Token_1.sol#162)

Recommended action: Review this detector result in context, validate its impact, implement a targeted fix where necessary, and add regression tests.

FINDING 21 | missing-zero-check

Low

Detector: missing-zero-check

Code location: Line 201, 213

Scanner description: D3Token.setMainPair(address).quoteToken_ (uploads/d52437ee22094038b7eb7b15a761c25f_D3Token_1.sol#201) lacks a zero-check on : - quoteToken = quoteToken_ (uploads/d52437ee22094038b7eb7b15a761c25f_D3Token_1.sol#213)

Recommended action: Review this detector result in context, validate its impact, implement a targeted fix where necessary, and add regression tests.

FINDING 22 | reentrancy-events

Low

Detector: reentrancy-events

Code location: Line 355, 356, 357, 358, 359, 360, 361, 362, 363, 364, 365, 366, 367, 368, 369, 370, 371, 372, 373, 374, 375, 376, 377

Scanner description: Reentrancy in D3Token.executeDeflation() (uploads/d52437ee22094038b7eb7b15a761c25f_D3Token_1.sol#355-377): External calls: - ID3Pair(mainPair).sync() (uploads/d52437ee22094038b7eb7b15a761c25f_D3Token_1.sol#374) Event emitted after the call(s): - DeflationExecuted(mainPair,msg.sender,pairBalanceBefore,burnAmount,block.timestamp) (uploads/d52437ee22094038b7eb7b15a761c25f_D3Token_1.sol#376)

Recommended action: Review this detector result in context, validate its impact, implement a targeted fix where necessary, and add regression tests.

FINDING 23 | timestamp

Low

Detector: timestamp

Code location: Line 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67

Scanner description: ERC20Permit.permit(address,address,uint256,uint256,uint8,bytes32,bytes32) (node_modules/@openzeppelin/contracts/token/ERC20/extensions/ERC20Permit.sol#44-67) uses timestamp for comparisons Dangerous comparisons: - block.timestamp > deadline (node_modules/@openzeppelin/contracts/token/ERC20/extensions/ERC20Permit.sol#53)

Recommended action: Review this detector result in context, validate its impact, implement a targeted fix where necessary, and add regression tests.

FINDING 24 | timestamp

Low

Detector: timestamp

Code location: Line 355, 356, 357, 358, 359, 360, 361, 362, 363, 364, 365, 366, 367, 368, 369, 370, 371, 372, 373, 374, 375, 376, 377

Scanner description: D3Token.executeDeflation() (uploads/d52437ee22094038b7eb7b15a761c25f_D3Token_1.sol#355-377) uses timestamp for comparisons Dangerous comparisons: - lastDeflationAt != 0 && block.timestamp < nextDeflationAt() (uploads/d52437ee22094038b7eb7b15a761c25f_D3Token_1.sol#358)

Recommended action: Review this detector result in context, validate its impact, implement a targeted fix where necessary, and add regression tests.

FINDING 25 | timestamp

Low

Detector: timestamp

Code location: Line 470, 471, 472, 473, 474, 475, 476, 477, 478, 479, 480, 481, 482, 483, 484, 485, 486, 487, 488, 489, 490, 491, 492

Scanner description: D3Token._currentPairState() (uploads/d52437ee22094038b7eb7b15a761c25f_D3Token_1.sol#470-492) uses timestamp for comparisons Dangerous comparisons: - pairTimestamp != currentTimestamp (uploads/d52437ee22094038b7eb7b15a761c25f_D3Token_1.sol#484)

Recommended action: Review this detector result in context, validate its impact, implement a targeted fix where necessary, and add regression tests.

FINDING 26 | timestamp

Low

Detector: timestamp

Code location: Line 301, 302, 303, 304, 305, 306, 307, 308

Scanner description: D3Token.sealTaxBps() (uploads/d52437ee22094038b7eb7b15a761c25f_D3Token_1.sol#301-308) uses timestamp for comparisons Dangerous comparisons: - liquidity < SEAL_POOL_LEVEL_1 (uploads/d52437ee22094038b7eb7b15a761c25f_D3Token_1.sol#303) - liquidity < SEAL_POOL_LEVEL_2 (uploads/d52437ee22094038b7eb7b15a761c25f_D3Token_1.sol#304) - liquidity < SEAL_POOL_LEVEL_3 (uploads/d52437ee22094038b7eb7b15a761c25f_D3Token_1.sol#305) - liquidity < SEAL_POOL_LEVEL_4 (uploads/d52437ee22094038b7eb7b15a761c25f_D3Token_1.sol#306)

Recommended action: Review this detector result in context, validate its impact, implement a targeted fix where necessary, and add regression tests.

FINDING 27 | timestamp

Low

Detector: timestamp

Code location: Line 287, 288, 289, 290, 291, 292, 293, 294, 295, 296, 297, 298, 299

Scanner description: D3Token.gateTaxBps() (uploads/d52437ee22094038b7eb7b15a761c25f_D3Token_1.sol#287-299) uses timestamp for comparisons Dangerous comparisons: - fastPrice == 0 || slowPrice == 0 || fastPrice >= slowPrice (uploads/d52437ee22094038b7eb7b15a761c25f_D3Token_1.sol#290) - deviationBps <= GATE_START_DEVIATION_BPS (uploads/d52437ee22094038b7eb7b15a761c25f_D3Token_1.sol#293) - deviationBps >= GATE_FULL_DEVIATION_BPS (uploads/d52437ee22094038b7eb7b15a761c25f_D3Token_1.sol#294)

Recommended action: Review this detector result in context, validate its impact, implement a targeted fix where necessary, and add regression tests.

FINDING 28 | timestamp

Low

Detector: timestamp

Code location: Line 311, 312, 313, 314, 315, 316, 317, 318, 319, 320, 321, 322, 323

Scanner description: D3Token.currentSellTaxBreakdown() (uploads/d52437ee22094038b7eb7b15a761c25f_D3Token_1.sol#311-323) uses timestamp for comparisons Dangerous comparisons: - totalTaxBps > MAX_TOTAL_SELL_TAX_BPS (uploads/d52437ee22094038b7eb7b15a761c25f_D3Token_1.sol#322)

Recommended action: Review this detector result in context, validate its impact, implement a targeted fix where necessary, and add regression tests.

FINDING 29 | timestamp

Low

Detector: timestamp

Code location: Line 417, 418, 419, 420, 421, 422, 423, 424, 425, 426, 427, 428, 429, 430, 431, 432, 433, 434, 435, 436, 437, 438, 439, 440, 441, 442, 443, 444, 445, 446, 447, 448, 449, 450, 451, 452, 453, 454, 455

Scanner description: D3Token._updateOracle() (uploads/d52437ee22094038b7eb7b15a761c25f_D3Token_1.sol#417-455) uses timestamp for comparisons Dangerous comparisons: - currentSpot == 0 (uploads/d52437ee22094038b7eb7b15a761c25f_D3Token_1.sol#421) - uint256(currentTimestamp) < uint256(latest.timestamp) + ORACLE_INTERVAL (uploads/d52437ee22094038b7eb7b15a761c25f_D3Token_1.sol#437)

Recommended action: Review this detector result in context, validate its impact, implement a targeted fix where necessary, and add regression tests.

FINDING 30 | timestamp

Low

Detector: timestamp

Code location: Line 495, 496, 497, 498, 499, 500, 501, 502, 503, 504, 505, 506, 507, 508, 509, 510, 511, 512, 513, 514, 515, 516, 517, 518, 519, 520, 521, 522, 523, 524, 525, 526, 527, 528, 529, 530, 531, 532, 533, 534

Scanner description: D3Token._computeTwap(uint256,bool) (uploads/d52437ee22094038b7eb7b15a761c25f_D3Token_1.sol#495-534) uses timestamp for comparisons Dangerous comparisons: - observation.timestamp <= targetTimestamp (uploads/d52437ee22094038b7eb7b15a761c25f_D3Token_1.sol#515) - referenceObservation.timestamp == 0 || currentTimestamp <= referenceObservation.timestamp (uploads/d52437ee22094038b7eb7b15a761c25f_D3Token_1.sol#524) - uint256(currentTimestamp) > window (uploads/d52437ee22094038b7eb7b15a761c25f_D3Token_1.sol#499-501)

Recommended action: Review this detector result in context, validate its impact, implement a targeted fix where necessary, and add regression tests.

FINDING 31 | timestamp

Low

Detector: timestamp

Code location: Line 551, 552, 553, 554, 555, 556, 557, 558, 559, 560, 561, 562, 563, 564, 565, 566, 567, 568, 569

Scanner description: D3Token._transferWithTax(address,address,uint256,uint256,uint256,uint256,bool) (uploads/d52437ee22094038b7eb7b15a761c25f_D3Token_1.sol#551-569) uses timestamp for comparisons Dangerous comparisons: - taxAmount > 0 (uploads/d52437ee22094038b7eb7b15a761c25f_D3Token_1.sol#563)

Recommended action: Review this detector result in context, validate its impact, implement a targeted fix where necessary, and add regression tests.

FINDING 32 | timestamp

Low

Detector: timestamp

Code location: Line 340, 341, 342, 343

Scanner description: D3Token.nextDeflationAt() (uploads/d52437ee22094038b7eb7b15a761c25f_D3Token_1.sol#340-343) uses timestamp for comparisons Dangerous comparisons: - lastDeflationAt == 0 (uploads/d52437ee22094038b7eb7b15a761c25f_D3Token_1.sol#341)

Recommended action: Review this detector result in context, validate its impact, implement a targeted fix where necessary, and add regression tests.

FINDING 33 | assembly

Informational

Detector: assembly

Code location: Line 89, 90, 91, 92, 93, 94

Scanner description: StorageSlot.getUint256Slot(bytes32) (node_modules/@openzeppelin/contracts/utils/StorageSlot.sol#89-94) uses assembly - INLINE ASM (node_modules/@openzeppelin/contracts/utils/StorageSlot.sol#91-93)

Recommended action: Review this detector result in context, validate its impact, implement a targeted fix where necessary, and add regression tests.

FINDING 34 | assembly

Informational

Detector: assembly

Code location: Line 30, 31, 32, 33, 34, 35, 36, 37

Scanner description: MessageHashUtils.toEthSignedMessageHash(bytes32) (node_modules/@openzeppelin/contracts/utils/cryptography/MessageHashUtils.sol#30-37) uses assembly - INLINE ASM (node_modules/@openzeppelin/contracts/utils/cryptography/MessageHashUtils.sol#32-36)

Recommended action: Review this detector result in context, validate its impact, implement a targeted fix where necessary, and add regression tests.

FINDING 35 | assembly

Informational

Detector: assembly

Code location: Line 59, 60, 61, 62, 63, 64

Scanner description: StorageSlot.getAddressSlot(bytes32) (node_modules/@openzeppelin/contracts/utils/StorageSlot.sol#59-64) uses assembly - INLINE ASM (node_modules/@openzeppelin/contracts/utils/StorageSlot.sol#61-63)

Recommended action: Review this detector result in context, validate its impact, implement a targeted fix where necessary, and add regression tests.

FINDING 36 | assembly

Informational

Detector: assembly

Code location: Line 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184, 185, 186, 187, 188, 189, 190, 191, 192, 193, 194, 195, 196, 197, 198, 199, 200, 201, 202

Scanner description: Math.mulDiv(uint256,uint256,uint256) (node_modules/@openzeppelin/contracts/utils/math/Math.sol#123-202) uses assembly - INLINE ASM (node_modules/@openzeppelin/contracts/utils/math/Math.sol#130-133) - INLINE ASM (node_modules/@openzeppelin/contracts/utils/math/Math.sol#154-161) - INLINE ASM (node_modules/@openzeppelin/contracts/utils/math/Math.sol#167-176)

Recommended action: Review this detector result in context, validate its impact, implement a targeted fix where necessary, and add regression tests.

FINDING 37 | assembly

Informational

Detector: assembly

Code location: Line 76, 77, 78, 79, 80, 81, 82, 83, 84, 85

Scanner description: MessageHashUtils.toTypedDataHash(bytes32,bytes32) (node_modules/@openzeppelin/contracts/utils/cryptography/MessageHashUtils.sol#76-85) uses assembly - INLINE ASM (node_modules/@openzeppelin/contracts/utils/cryptography/MessageHashUtils.sol#78-84)

Recommended action: Review this detector result in context, validate its impact, implement a targeted fix where necessary, and add regression tests.

FINDING 38 | assembly

Informational

Detector: assembly

Code location: Line 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44

Scanner description: Strings.toString(uint256) (node_modules/@openzeppelin/contracts/utils/Strings.sol#24-44) uses assembly - INLINE ASM (node_modules/@openzeppelin/contracts/utils/Strings.sol#30-32) - INLINE ASM (node_modules/@openzeppelin/contracts/utils/Strings.sol#36-38)

Recommended action: Review this detector result in context, validate its impact, implement a targeted fix where necessary, and add regression tests.

FINDING 39 | assembly

Informational

Detector: assembly

Code location: Line 69, 70, 71, 72, 73, 74

Scanner description: StorageSlot.getBooleanSlot(bytes32) (node_modules/@openzeppelin/contracts/utils/StorageSlot.sol#69-74) uses assembly - INLINE ASM (node_modules/@openzeppelin/contracts/utils/StorageSlot.sol#71-73)

Recommended action: Review this detector result in context, validate its impact, implement a targeted fix where necessary, and add regression tests.

FINDING 40 | assembly

Informational

Detector: assembly

Code location: Line 119, 120, 121, 122, 123, 124

Scanner description: StorageSlot.getBytesSlot(bytes32) (node_modules/@openzeppelin/contracts/utils/StorageSlot.sol#119-124) uses assembly - INLINE ASM (node_modules/@openzeppelin/contracts/utils/StorageSlot.sol#121-123)

Recommended action: Review this detector result in context, validate its impact, implement a targeted fix where necessary, and add regression tests.

FINDING 41 | assembly

Informational

Detector: assembly

Code location: Line 99, 100, 101, 102, 103, 104

Scanner description: StorageSlot.getStringSlot(bytes32) (node_modules/@openzeppelin/contracts/utils/StorageSlot.sol#99-104) uses assembly - INLINE ASM (node_modules/@openzeppelin/contracts/utils/StorageSlot.sol#101-103)

Recommended action: Review this detector result in context, validate its impact, implement a targeted fix where necessary, and add regression tests.

FINDING 42 | assembly

Informational

Detector: assembly

Code location: Line 129, 130, 131, 132, 133, 134

Scanner description: StorageSlot.getBytesSlot(bytes) (node_modules/@openzeppelin/contracts/utils/StorageSlot.sol#129-134) uses assembly - INLINE ASM (node_modules/@openzeppelin/contracts/utils/StorageSlot.sol#131-133)

Recommended action: Review this detector result in context, validate its impact, implement a targeted fix where necessary, and add regression tests.

FINDING 43 | assembly

Informational

Detector: assembly

Code location: Line 109, 110, 111, 112, 113, 114

Scanner description: StorageSlot.getStringSlot(string) (node_modules/@openzeppelin/contracts/utils/StorageSlot.sol#109-114) uses assembly - INLINE ASM (node_modules/@openzeppelin/contracts/utils/StorageSlot.sol#111-113)

Recommended action: Review this detector result in context, validate its impact, implement a targeted fix where necessary, and add regression tests.

FINDING 44 | assembly

Informational

Detector: assembly

Code location: Line 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73

Scanner description: ShortStrings.toString(ShortString) (node_modules/@openzeppelin/contracts/utils/ShortStrings.sol#63-73) uses assembly - INLINE ASM (node_modules/@openzeppelin/contracts/utils/ShortStrings.sol#68-71)

Recommended action: Review this detector result in context, validate its impact, implement a targeted fix where necessary, and add regression tests.

FINDING 45 | assembly

Informational

Detector: assembly

Code location: Line 79, 80, 81, 82, 83, 84

Scanner description: StorageSlot.getBytes32Slot(bytes32) (node_modules/@openzeppelin/contracts/utils/StorageSlot.sol#79-84) uses assembly - INLINE ASM (node_modules/@openzeppelin/contracts/utils/StorageSlot.sol#81-83)

Recommended action: Review this detector result in context, validate its impact, implement a targeted fix where necessary, and add regression tests.

FINDING 46 | assembly

Informational

Detector: assembly

Code location: Line 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73

Scanner description: ECDSA.tryRecover(bytes32,bytes) (node_modules/@openzeppelin/contracts/utils/cryptography/ECDSA.sol#56-73) uses assembly - INLINE ASM (node_modules/@openzeppelin/contracts/utils/cryptography/ECDSA.sol#64-68)

Recommended action: Review this detector result in context, validate its impact, implement a targeted fix where necessary, and add regression tests.

FINDING 47 | dead-code

Informational

Detector: dead-code

Code location: Line 170, 171, 172, 173, 174

Scanner description: AccessControl._setRoleAdmin(bytes32,bytes32) (node_modules/@openzeppelin/contracts/access/AccessControl.sol#170-174) is never used and should be removed

Recommended action: Review this detector result in context, validate its impact, implement a targeted fix where necessary, and add regression tests.

FINDING 48 | dead-code

Informational

Detector: dead-code

Code location: Line 25, 26, 27

Scanner description: Context._contextSuffixLength() (node_modules/@openzeppelin/contracts/utils/Context.sol#25-27) is never used and should be removed

Recommended action: Review this detector result in context, validate its impact, implement a targeted fix where necessary, and add regression tests.

FINDING 49 | dead-code

Informational

Detector: dead-code

Code location: Line 40, 41, 42, 43, 44, 45

Scanner description: Nonces._useCheckedNonce(address,uint256) (node_modules/@openzeppelin/contracts/utils/Nonces.sol#40-45) is never used and should be removed

Recommended action: Review this detector result in context, validate its impact, implement a targeted fix where necessary, and add regression tests.

FINDING 50 | dead-code

Informational

Detector: dead-code

Code location: Line 21, 22, 23

Scanner description: Context._msgData() (node_modules/@openzeppelin/contracts/utils/Context.sol#21-23) is never used and should be removed

Recommended action: Review this detector result in context, validate its impact, implement a targeted fix where necessary, and add regression tests.

FINDING 51 | solc-version

Informational

Detector: solc-version

Code location: Line 4, 3, 5, 2

Scanner description: Version constraint ^0.8.20 contains known severe issues (https://solidity.readthedocs.io/en/latest/bugs.html) - VerbatimInvalidDeduplication - FullInlinerNonExpressionSplitArgumentEvaluationOrder - MissingSideEffectsOnSelectorAccess. It is used by: - ^0.8.20 (node_modules/@openzeppelin/contracts/access/AccessControl.sol#4) - ^0.8.20 (node_modules/@openzeppelin/contracts/access/IAccessControl.sol#4) - ^0.8.20 (node_modules/@openzeppelin/contracts/access/Ownable.sol#4) - ^0.8.20 (node_modules/@openzeppelin/contracts/interfaces/IERC5267.sol#4) - ^0.8.20 (node_modules/@openzeppelin/contracts/interfaces/draft-IERC6093.sol#3) - ^0.8.20 (node_modules/@openzeppelin/contracts/token/ERC20/ERC20.sol#4) - ^0.8.20 (node_modules/@openzeppelin/contracts/token/ERC20/IERC20.sol#4) - ^0.8.20 (node_modules/@openzeppelin/contracts/token/ERC20/extensions/ERC20Burnable.sol#4) - ^0.8.20 (node_modules/@openzeppelin/contracts/token/ERC20/extensions/ERC20Permit.sol#4) - ^0.8.20 (node_modules/@openzeppelin/contracts/token/ERC20/extensions/IERC20Metadata.sol#4) - ^0.8.20 (node_modules/@openzeppelin/contracts/token/ERC20/extensions/IERC20Permit.sol#4) - ^0.8.20 (node_modules/@openzeppelin/contracts/utils/Context.sol#4) - ^0.8.20 (node_modules/@openzeppelin/contracts/utils/Nonces.sol#3) - ^0.8.20 (node_modules/@openzeppelin/contracts/utils/ShortStrings.sol#4) - ^0.8.20 (node_modules/@openzeppelin/contracts/utils/StorageSlot.sol#5) - ^0.8.20 (node_modules/@openzeppelin/contracts/utils/Strings.sol#4) - ^0.8.20 (node_modules/@openzeppelin/contracts/utils/cryptography/ECDSA.sol#4) - ^0.8.20 (node_modules/@openzeppelin/contracts/utils/cryptography/EIP712.sol#4) - ^0.8.20 (node_modules/@openzeppelin/contracts/utils/cryptography/MessageHashUtils.sol#4) - ^0.8.20 (node_modules/@openzeppelin/contracts/utils/introspection/ERC165.sol#4) - ^0.8.20 (node_modules/@openzeppelin/contracts/utils/introspection/IERC165.sol#4) - ^0.8.20 (node_modules/@openzeppelin/contracts/utils/math/Math.sol#4) - ^0.8.20 (node_modules/@openzeppelin/contracts/utils/math/SignedMath.sol#4) - ^0.8.20 (uploads/d52437ee22094038b7eb7b15a761c25f_D3Token_1.sol#2)

Recommended action: Verify the compiler version and applicable compiler bug conditions. Select a patched compiler version compatible with the project, rebuild the contract, and re-run the checks.

FINDING 52 | naming-convention

Informational

Detector: naming-convention

Code location: Line 157, 158, 159

Scanner description: Function EIP712._EIP712Version() (node_modules/@openzeppelin/contracts/utils/cryptography/EIP712.sol#157-159) is not in mixedCase

Recommended action: Review this detector result in context, validate its impact, implement a targeted fix where necessary, and add regression tests.

FINDING 53 | naming-convention

Informational

Detector: naming-convention

Code location: Line 89

Scanner description: Function IERC20Permit.DOMAIN_SEPARATOR() (node_modules/@openzeppelin/contracts/token/ERC20/extensions/IERC20Permit.sol#89) is not in mixedCase

Recommended action: Review this detector result in context, validate its impact, implement a targeted fix where necessary, and add regression tests.

FINDING 54 | naming-convention

Informational

Detector: naming-convention

Code location: Line 80, 81, 82

Scanner description: Function ERC20Permit.DOMAIN_SEPARATOR() (node_modules/@openzeppelin/contracts/token/ERC20/extensions/ERC20Permit.sol#80-82) is not in mixedCase

Recommended action: Review this detector result in context, validate its impact, implement a targeted fix where necessary, and add regression tests.

FINDING 55 | naming-convention

Informational

Detector: naming-convention

Code location: Line 146, 147, 148

Scanner description: Function EIP712._EIP712Name() (node_modules/@openzeppelin/contracts/utils/cryptography/EIP712.sol#146-148) is not in mixedCase

Recommended action: Review this detector result in context, validate its impact, implement a targeted fix where necessary, and add regression tests.

FINDING 56 | too-many-digits

Informational

Detector: too-many-digits

Code location: Line 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123

Scanner description: ShortStrings.slitherConstructorConstantVariables() (node_modules/@openzeppelin/contracts/utils/ShortStrings.sol#40-123) uses literals with too many digits: -FALLBACK_SENTINEL = 0x00000000000000000000000000000000000000000000000000000000000000FF (node_modules/@openzeppelin/contracts/utils/ShortStrings.sol#42)

Recommended action: Review this detector result in context, validate its impact, implement a targeted fix where necessary, and add regression tests.

FINDING 57 | constable-states

Informational

Detector: constable-states

Code location: Line 52

Scanner description: EIP712._versionFallback (node_modules/@openzeppelin/contracts/utils/cryptography/EIP712.sol#52) should be constant

Recommended action: Review this detector result in context, validate its impact, implement a targeted fix where necessary, and add regression tests.

FINDING 58 | constable-states

Informational

Detector: constable-states

Code location: Line 51

Scanner description: EIP712._nameFallback (node_modules/@openzeppelin/contracts/utils/cryptography/EIP712.sol#51) should be constant

Recommended action: Review this detector result in context, validate its impact, implement a targeted fix where necessary, and add regression tests.

SECURITY ASSESSMENT

CONCLUSION

Motech Audit has completed its security review of the revised D3Token.sol smart contract. Based on the reviewed source code, the revised implementation demonstrates meaningful improvements in its security control design, particularly in addressing previously identified honeypot-related concerns.

The removal of the emergency withdrawal functions reduces the scope for privileged asset extraction through those specific mechanisms, while the removal of the dedicated sell-disable mechanism improves the transparency of the contract's trading controls. The redesigned deflation mechanism introduces a fixed 0.2% burn rate, a once-per-UTC-day execution limit, direct token burning to the zero address, and liquidity-pair synchronization following execution. These changes provide a more clearly defined approach to token supply reduction and reduce reliance on the previous recipient-distribution mechanism.

The contract also incorporates a fixed initial supply model, configurable transaction taxation, TWAP-related price monitoring, and liquidity-pair interaction logic. These mechanisms form part of the contract's intended tokenomics and trading design. The project team reports that 24 local Hardhat simulation scenarios passed, covering key trading behaviors, access-control boundaries, deflation execution, token supply reduction, liquidity-pair synchronization, and a simulated arbitrage scenario. These results have been considered as project-provided testing evidence in this assessment.

Despite these improvements, certain risks remain. The owner retains the ability to modify tax-exempt addresses and change the configured main liquidity pair, while the owner or administrator can disable the deflation mechanism. The maximum total sell tax of 30% may also materially affect users' exit costs. In addition, the interaction between tax exemptions, dynamic sell taxation, oracle observations, liquidity conditions, and privileged configuration should be considered when evaluating the contract's operational security and user protection.

SECURITY ASSESSMENT

CONCLUSION

Final Assessment: The revised D3Token source code demonstrates material remediation of the previously identified emergency withdrawal and sell-disable mechanisms. The updated deflation design also provides a more explicit and structured token-burning process. These changes represent positive improvements to the contract's security control design. However, the remaining administrative privileges, tax configuration powers, liquidity-pair management, and token distribution characteristics continue to contribute to the contract's overall risk profile. The removal of the previously identified mechanisms should not be interpreted as proof that all potential vulnerabilities, manipulation scenarios, or economic risks have been eliminated.

Motech Audit recommends appropriate disclosure and management of privileged controls, together with continued testing of trading restrictions, tax-exemption interactions, oracle behavior, liquidity-pair changes, and adversarial transaction scenarios. The final assessment applies to the specific source file reviewed and the information available within the audit scope.

Disclaimer: This audit evaluates the specified smart contract within the defined review scope. It does not provide any representation or warranty regarding the project's business model, investment attractiveness, future token price, or long-term code sustainability. This report does not cover other contracts deployed by the owner unless they are explicitly included in the audit scope. No smart contract audit can guarantee the complete absence of vulnerabilities or future losses.